

Syllabus for ENGR-E 599: Topics in Intelligent Systems Engineering, Radiation Effects and Reliability of Microelectronics

Section: 11750

Course Description:

This course introduces the space radiation environment and its effects on microelectronics. The basic mechanisms of cumulative and transient radiation effects in state-of-the-art semiconductor devices are discussed, with special focus on the challenges of modeling, simulation, and measurement. Further, the implications of technology scaling are considered. Students will be exposed to industry-standard tools and techniques for estimating the reliability of microelectronics in extreme space environments.

Course Pre/Co-Requisites: None

Meeting Dates: Tuesday / Thursday, 12:45 pm-2:00 pm Luddy Hall (IF) Room 4063

Course delivery method: in-person

Learning Outcomes:

- Identify, formulate, and solve engineering problems related to the use of microelectronics in space environments
- Demonstrate an ability to communicate effectively
- Recognize the need for, and an ability to engage in life-long learning
- Demonstrate the use of techniques, skills, and modern tools for analyzing microelectronics radiation effects and reliability degradation

Assessment:	Assignments	20%
	Midterm exam	20%
	Mini design project	40%
	Review article	20%

Major due dates:

Note to students: these dates are subject to change with adequate notification via Canvas.

- Homework assignments: due most Thursdays
- Midterm exam: Tuesday, Oct. 20 (in lecture)
- Review paper: Thursday, Nov. 19
- Final project deadline: Thursday, Dec. 17, 2:40 pm (regular final exam time).

Required materials:

- There is no required textbook; all course reference materials are supplied in-class and on the course [GitHub](#) page.
- No software purchases are required; the course uses free open-source software such as [SPENVIS](#), [CRÉME](#), [nanoHUB](#), and [RADHUB](#). All tool suites require individual registration.
- Students are expected to supply a personal working laptop computer with permissions to install and use course software.

Other considerations: Not applicable.